

Attachment Rescue: giving NCI's buried supplementary datasets a permanent home

NCI Office of Data Sharing Impact Prize · Track 1: Research Output Sharing and Reuse Ideas · Submitted by Ochiai, Ojira · Individual

Prompt 1: Significance and Approach

The gap. Before the 2023 NIH Data Management and Sharing Policy, the normal way to share a cancer dataset was to attach it to the paper: gene expression tables, patient cohort tables, drug screening results and mass spectrometry measurements, saved as Excel files and uploaded as "Supplementary Data 3." Those files still exist; PubMed Central preserves them. But no dataset catalogue lists them, no search indexes their contents, and nothing describes what is inside. A researcher looking for glioblastoma cohort data with survival and IDH status cannot find a 121-patient table that has sat in a supplementary file since 2018 without already knowing the paper.

How big it is. We measured it. Europe PMC lists 58,469 NCI-funded open-access papers from 2016 to 2022; 26,228 have supplementary files and no text-mined accession for any of ten data repositories (GEO, SRA, dbGaP, EGA, ArrayExpress, PRIDE, BioProject, MetaboLights, PDB, GenBank). We drew random samples of 300 papers for each of four cancers (glioblastoma, breast, colorectal, leukemia), downloaded every attachment, and opened every spreadsheet, CSV, Word and PDF file. A file counts as a dataset only if its largest table has at least 50 rows and 3 columns, or at least 1,000 rows. **The four cancers agree: 19.6%, 18.8%, 19.4% and 17.2% of papers contain at least one genuine data table; 15.1%, 15.1%, 15.0% and 12.6% hold it as a native spreadsheet rather than a table inside a PDF. Pooled, 202 of 1,076 papers (19%) hold a data table and 156 (14.5%, confidence interval 12% to 17%) a native spreadsheet.** 167 of the 202 (83%) are CC BY or CCo, so their files can be re-deposited today. Tables ran from 50 rows to 152,683. A person opened 105 of the machine's decisions: all 60 files it called datasets were real data tables, and every table it had missed was under 50 rows. The verdicts are published file by file. We then ran a census of the full population by reading each paper's full text for its list of attachments, downloading nothing. Across 25,138 readable papers there are 73,795 attachments. **4,759 papers (18.9%) carry at least one spreadsheet, CSV or tab-delimited text attachment, 15,458 such files in total, 15.9 gigabytes of Excel alone.** 3,824 of those papers (80%) are licensed CC BY or CCo. Measured the same way, by filename, the samples give 17.9%; the census and the samples agree within a point. We also tested our own filter: in a random 500 population papers, 15% did point to a repository somewhere the accession miner had not caught, and 13.5% of spreadsheet papers already have a publisher-assigned Figshare identifier, verified file by file. Netting those out and applying the samples' rates, the honest estimate is roughly 2,800 to 3,900 papers

holding 8,000 to 11,000 buried dataset files, about three-quarters under a license that permits re-deposit. About 4% of papers could not be read and are excluded.

Who is affected. Researchers, because data they need is unfindable; trainees and small labs most of all, because they lack the network that knows which paper to open. Students, patient advocates and NCI's own Data Jamboree participants, who are told computing skills are not required and then face a literature that hides its data in attachments. The original authors, whose data earns no citations because it has no identifier. And NCI, which paid for these datasets twice and receives no reuse in return.

Why it matters to NCI and ODS goals. These papers were compliant. They followed the norms of their time, and the data still vanished from view. That is the exact situation the challenge describes when it says policy alone does not drive culture change. It also bears on NCI's commitment to broad, rapid and equitable sharing: data only insiders can find is not broadly shared. ODS's own workshop report names "incomplete, incorrect, or missing metadata" and the findability of siloed data as major bottlenecks; its Data Jamboree paper names "limited awareness of available datasets" as a primary barrier. A supplementary file is the purest case of both: real data with no metadata, in a silo of one.

The approach. A retrospective rescue, run over public infrastructure that already exists:

1. **Find.** Query Europe PMC for NCI-funded open-access papers with supplementary files and no repository accession, and read each paper's full text to list its attachments. One call per paper, no downloads.
2. **Sort.** Download the tabular attachments and classify each: dataset, small table, or not data. Separate primary measurements (cohort tables, expression tables, counts, spectra) from derived results (gene lists, pathway tables). Skip peer-review files, reporting summaries and figures automatically.
3. **Describe.** Generate a standard dataset record for each, in the DataCite format repositories use: title, creators and license from the article, article DOI and PMC identifier, funder, format, size, sheet names, and a one-sentence plain-language description of what the table contains. A person spot-checks.
4. **Deposit or register.** For CC BY and CCo articles, deposit the unmodified file in one of the free general-purpose repositories NIH already recognizes, which assigns it a permanent identifier (a DOI). For no-derivatives licenses, and for any file whose authors object, register a metadata record that points to the file's existing location in PubMed Central instead of copying it. Who deposits, and how authors are told, is addressed under constraints below.
5. **List.** Deliver the rescued records to the Index of NCI Studies as a curated source file in the same format its pipeline already accepts from other curated sources, so the datasets appear beside GEO and dbGaP records in NCI's own catalogue.

6. **Keep alive.** Check every rescued link and identifier monthly and publish the result, so the collection cannot quietly rot.

Steps 1 through 3 have run on 1,200 papers across four cancers, and step 4 as far as packaging: eight glioblastoma datasets sit ready for deposit with complete metadata, awaiting the governance decision below. The proposal is to run the whole pipeline over the full NCI corpus, cancer by cancer.

Prompt 2: Potential Impact on the Cancer Research Community

Immediate. Thousands of existing cancer datasets become findable, citable and described. The NCI open-access corpus from 2016 to 2022 holds 15,458 spreadsheet and CSV attachments in papers that cite no repository. By rates measured in four cancer types, on the order of 8,000 to 11,000 of them are buried datasets, about three-quarters freely re-depositable. None of this requires new data; the datasets exist today, and the cost of finding them is all that stands between them and reuse.

Breadth. The gap is not specific to any cancer, any NCI division, or cancer itself; attaching data to papers was standard practice across biomedicine before 2023. The pipeline is defined by a funder tag and a date range, so any NCI Division, Office or Center, or any other NIH institute, can run it unchanged. The four pilot cancers agree within three points, which suggests the rate reflects publishing practice rather than any one field.

For non-data scientists. A rescued dataset arrives with a plain-language description and a stable identifier. A community college instructor can hand students a real patient cohort table with a DOI. A Data Jamboree team can search the catalogue for "cohort with survival and MGMT status" and get a hit. We propose offering the rescued collection to the November 2026 NCI Data Jamboree as a ready-made resource, putting rescued data in front of exactly the non-traditional users the challenge names, at an event ODS runs.

For authors and for credit. Each rescued dataset gains an identifier linked to its article, so invisible data can start to accrue citations. Papers whose data statements link to a repository receive up to 25% more citations than those without (Colavizza et al., 2020); the relationship is correlational, but rescue gives authors that identifier at no effort to them.

For policy and infrastructure across NCI. The census gives ODS something it does not currently have: a measured, per-year, per-journal account of how much NCI-funded data was shared only as attachments and how much is legally recoverable. That informs data-sharing guidance, lets program officers report the public benefit of past investments concretely, and gives the Index of NCI Studies its first source of datasets drawn from publications rather than repositories.

Prompt 3: Innovation and Awareness of Existing Efforts

What exists. We looked hard for prior work, and there is some.

- **Large-scale extraction of supplementary files has been done.** Gobeill and colleagues (Bioinformatics Advances, 2025) processed 36.2 million supplementary files from 6.2 million PubMed Central articles, found that 97% of database accession numbers appear only in supplements, and built a text search index; the Europe PMC team (Kafkas et al., 2015) mined supplements for accession numbers earlier. Both treat supplements as text to search, not datasets to catalogue; neither classifies files as data, produces dataset metadata, assigns identifiers, or lists anything in a discovery catalogue.
- **Publishers mirror new supplements prospectively.** PLOS assigns a component DOI to every supporting information file and mirrors it to Figshare; Springer Nature and BMC auto-deposit supplements to Figshare for hundreds of journals. These are forward-looking, publisher-scoped, and register the files as article components, not as described datasets. We checked our census against Figshare: files from PLOS and BMC-family journals, 13.5% of census spreadsheet papers when verified file by file, already have a Figshare identifier; files from Nature Communications, Scientific Reports, eLife, Oncotarget and Cell Press journals, the large majority, do not. Even the mirrored files have no dataset description and do not reach NCI's index, so they need the sorting, describing and listing steps but not the deposit step.
- **The scale of the problem is documented.** Federer et al. (2018) found 69% of PLOS ONE data availability statements pointed to the paper or its supplements and only 18% to a repository; Colavizza et al. (2020) found repository links in 21% of PLOS and 12% of BMC papers. Federer (2022) found DOIs stay resolvable far more often than plain links, which is why rescued files get one.
- **The catalogue side exists.** The Index of NCI Studies, managed by ODS and built by CBIIT, ingests datasets from GEO, SRA, dbGaP and several curated NCI sources through a public pipeline. It does not currently ingest anything from general-purpose repositories or from publications. But the INS team already runs automated enrichment over its records, including a published tool that maps study text to standard disease codes, so a curated source of publication-derived datasets is a next step for a pipeline built to take curated sources, not a new capability. NIH's Generalist Repository Ecosystem Initiative, which coordinates seven general-purpose repositories on metadata and identifiers, has no retrospective harvest component.

What is new. The combination, and the direction. We found no effort that has taken a funder's own back catalogue, classified which attachments are datasets, described them, given them identifiers, and delivered them into the funder's own catalogue. Extraction exists; deposit and listing do not. Prospective mirroring exists; retrospective rescue does not. The method is deliberately unoriginal: standard interfaces, standard table parsing, a standard metadata schema, existing repositories, an existing ingest pipeline. What is new is pointing it backwards at data shared in good faith and lost anyway.

What we are not proposing. Not a new repository, catalogue or metadata standard, and not a replacement for the 2023 policy. Rescue is the backfill for everything published before it.

Prompt 4: Transferability, Sustainability, and Feasibility

Path to adoption. Every component is public and already operated by someone else: Europe PMC serves the papers and attachments, general-purpose repositories assign identifiers for free, and the Index of NCI Studies pipeline is open source and ingests curated source files. Adoption means one new curated source in that pipeline, a decision inside NCI rather than new infrastructure.

What it would take. Seconds of compute and cents of storage per paper. The census needs no downloads; the rescue step for a cancer the size of glioblastoma is a few hundred files. Human time goes to spot-checking metadata and to one conversation: agreeing the source file format with the INS team, whose format is already published.

Constraints, stated plainly.

- *Licensing.* 80% of the spreadsheet-bearing papers are CC BY or CCo and their files can be deposited unchanged. About 10% are CC BY-NC-ND, which forbids adaptation; those datasets get a metadata record pointing at PubMed Central rather than a copy. Author manuscripts outside the open-access subset are excluded entirely. The pipeline enforces this by license tag before anything is copied.
- *Who deposits, and author consent.* We do not propose that a third party silently uploads authors' files. Two options, in order of preference: NCI or ODS acts as depositor under an institutional repository account, as NIH already does for NIH-authored manuscripts in PubMed Central; or, for a first release, authors are notified 60 days before deposit with a one-click opt-out, and files whose authors object receive a metadata-only record. Either way the record names the original authors as creators, links the article, carries a route for correction or removal, and NCI never owns the data. Tables describing human participants, even de-identified, get a review step before any copy is made.
- *Signal versus noise.* Most attachments are figures and methods. In our samples about two-thirds of papers had no tabular data at all. The classifier exists to keep that noise out, and its accuracy is published with the results. About half of rescued tables are derived results rather than raw measurements, and each record says which it is.
- *Duplication.* Some attachments duplicate data deposited elsewhere, and some already carry a publisher-assigned Figshare identifier. Records are matched against existing repository entries by article DOI and filename before deposit, and an existing identifier is reused, not duplicated.

- *The INS intake.* This is the one dependency on a decision by others. If INS does not add the source, the datasets still exist with DOIs in repositories NIH already recognizes, and the census still stands.

Why it persists. After the backlog, the pipeline runs quarterly on new open-access papers and monthly on link health, both as scheduled checks. What outlives any single team is the curated source file and the census, both public and reproducible.

Use of Generative AI

Generative AI (Anthropic's Claude) was used in preparing this submission: to help write and run the analysis code, to help draft and edit this narrative, and to draft the plain-language dataset descriptions in the example packages. The sampling, classification and counts were produced by scripts that are included with the supporting materials and can be rerun from public data. The classification was checked by hand on 105 files. All decisions about scope, claims and recommendations are the submitter's own, and the submitter takes full responsibility for the accuracy of the content.